TRACE-GATE: Auditing AI-Scientist Packages with Non-Directive Claim–Result–Code Checks

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Abstract

Long-horizon AI scientists can produce complete-looking research packages faster than those packages can be checked, making evidence inspection a practical bottleneck. We introduce TRACE-GATE, a paired generated benchmark and non-directive checking protocol that separates paper-only review, raw-artifact access, structured claim–result–code provenance, and a visible deterministic gate report. Across 36 clean or fault-injected packages, two frozen reviewer configurations, and 288 decisions, faulty-package acceptance was 77.8%, 11.1%, 5.6%, and 2.8% in the four conditions, while clean acceptance was 100%, 100%, 97.2%, and 100%. Most of the observed reduction came from raw-artifact access; the gate's increment beyond structured provenance was only -2.8 percentage points (case-clustered 95% interval $[-8.3, 0.0]$). Because 20 provider attempts were needed to obtain 16 analyzable batches after four outcome-blind schema exclusions, all effects are exploratory, but the decomposition shows why provenance gates should be evaluated as inspectable evidence rather than scientific-validity oracles.

1. Introduction

AI-scientist systems increasingly automate planning, execution, and writing. A remaining failure mode is more mundane: a plausible paper can disagree with its result, code, or provenance while still looking reviewable. Deliberately flawed manuscripts have historically survived expert review (Baxt et al., 1998), while independent execution can expose discrepancies at nontrivial cost (Nüst and Eglen, 2021). Recent agent benchmarks target computational reproduction (Siegel et al., 2025; Hu et al., 2025) and artifact evaluation (Baek and Pradel, 2026; Wu et al., 2026), but usually do not estimate false acceptance on paired clean and faulty research packages.

Machine-readable provenance addresses navigation, not truth. PROV-O represents entities and derivations (Lebo et al., 2013), and RO-Crate packages linked research objects (Soiland-Reyes et al., 2022); neither implies that a scientific claim is warranted. Conversely, defining acceptance as "reviewer accepts AND checker passes" makes a gate look effective by construction. We therefore ask a narrower question: how do raw artifacts, structured provenance, and a visible but non-directive deterministic report change a reviewer's own decision?

Our contribution is a controlled C0–C3 decomposition on a frozen paired fixture. It jointly reports faulty acceptance, clean acceptance, uncertainty, and review cost. The central result is deliberately asymmetric: artifact access accounts for most of the observed gain, whereas the increments from structure and gate evidence are small and unresolved.

2. TRACE-GATE Study

Generated packages. We programmatically created 18 lineages, each with one clean package and one single-fault partner (36 instances total). A package contains an anonymous short claim, contract, execution metadata, result, citation record, and claim–result–execution edges. Faults cover result direction, metric mismatch, stale artifacts, execution hashes, contract drift, negative-result omission, claim ceilings, code invariants, unsupported mechanisms, and citation entailment. The first 14 faulty partners map to deterministic predicates; two mechanism-support and two citation-entailment faults are explicitly semantic or out of scope. On this constructed census, the checker passed all 18 clean packages, failed all 14 mapped faults with the designated rule, and passed the four out-of-scope faults. These are fixture-conformance facts, not evidence of coverage on natural errors.

Single-fault construction. Each clean package was instantiated from the same bounded-study template, then assigned opaque case and pair identifiers. A corruption recipe

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Table 1. Exploratory outcomes on the frozen generated benchmark. FAR is faulty-package acceptance. Costs are per item; each item shared an 18-item call context.

Condition	Visible evidence	FAR ↓	Clean acc. ↑	Latency (s)	Input tok.	Output tok.
C0	Paper only	0.778	1.000	2.010	755.25	91.25
C1	Paper + raw artifacts	0.111	1.000	1.491	1061.75	81.75
C2	Structured provenance	0.056	0.972	1.745	1047.25	95.75
C3	C2 + non-directive gate report	0.028	1.000	1.622	1266.00	87.75

Table 2. Frozen gate registry and number of faulty partners. TG-09/10 deliberately return *out of scope*.

Rule	Checked relation	Mapped	Scope
TG-01	claim/result direction	2	mech.
TG-02	metric identity	2	mech.
TG-03	lifecycle/provenance edge	2	mech.
TG-04	execution/result digest	2	mech.
TG-05	frozen contract fields	2	mech.
TG-06	negative-result ledger	2	mech.
TG-07	claim ceiling	1	mech.
TG-08	code invariant	1	mech.
TG-09	mechanism support	2	semantic
TG-10	citation entailment	2	semantic

changed exactly one declared field or relation while preserving the remaining paper, contract, execution, result, citation, and provenance objects. The hidden manifest records the pair, transformation, gold label, fault stratum, package hash, and gate-report hash. Clean and faulty package hashes were frozen before review. This construction makes within-lineage contrasts interpretable, but it also aligns the 14 mechanical faults with the registry used to generate them.

Information conditions. C0 exposes only the anonymous paper. C1 adds raw code, logs, and results. C2 instead presents the package as structured claim–result–code provenance. C3 adds the deterministic TRACE-GATE report to C2. The report labels TG-01–TG-08 predicates PASS/FAIL and TG-09–TG-10 as out of scope. It is evidence only: the measured endpoint is always the reviewer's returned `accepted` Boolean; no FAIL vetoes or recodes it.

Review schedule and leakage controls. Two frozen configurations (`gpt-5.6-sol/max` and `gpt-5.4-mini/low`) each made eight ephemeral calls: four conditions crossed with two complementary shards. Every call contained 18 items from one condition, exactly one variant per lineage, and a 9/9 clean–fault balance. Thus paired counterparts never shared a call, but 18 items did share context. Gold labels and fault types were absent from visible packets and joined after inference. There was no human adjudication.

Blinded collection and integrity checks. Every visible item carried only a 20-character hash identifier and condition-appropriate materials. The common reviewer instruction asked whether the bounded central claim was adequately supported and explicitly treated automated output as evidence rather than a verdict. A strict schema required one identifier-preserving Boolean decision, confidence, an empty pre-join fault list, and a short reason per item. Before inference, canonical visible JSON was scanned for gold, condition, and corruption-manifest markers; every batch was checked for unique case and pair IDs. After inference, the runner required the returned identifier set to match the 18 inputs exactly, rejected duplicates or omissions, and only then joined case, condition, gold, and fault type. A final typed-table gate required 288 rows, four conditions for every reviewer–case key, stable labels across conditions, both frozen reviewer names, and the frozen contract and manifest hashes. Raw attempt logs and successful batch results were retained separately.

Endpoints and analysis. Let $Y_{imc} \in \{0, 1\}$ be acceptance for case i, reviewer configuration m, and condition c, and let G_i mark a faulty case. With two reviewers and 18 cases in each gold class,

$$\begin{aligned} \mathrm{FAR}_c &= \frac{1}{36} \sum_{i:G_i=1} \sum_m Y_{imc}, \\ \mathrm{CAR}_c &= \frac{1}{36} \sum_{i:G_i=0} \sum_m Y_{imc}. \end{aligned} \tag{1}$$

We report treatment-minus-baseline paired differences. Discordant faulty reviewer–case decisions give improvement/regression counts and a two-sided exact McNemar value. Percentile intervals use 2,000 paired bootstrap samples (seed 1729), resampling case IDs and retaining both reviewer rows per sampled case. The clustered interval is primary because the two reviewer rows on one package are dependent; McNemar values are descriptive. Latency is isolated batch wall time divided by 18, while input/output tokens are measured turn totals allocated in the same way. No missing decision was recoded as a rejection.

3. Results

Table 1 shows a 66.7-point FAR decrease from C0 to C1 with no clean-acceptance change. C3 versus C0 decreased FAR by 75.0 points (95% case-clustered interval

Table 3. Paired treatment-minus-baseline differences. Intervals cluster by package case.

Contrast	ΔFAR	95% interval	Imp./reg.
C1–C0	-0.667	$[-0.861, -0.472]$	24/0
C2–C1	-0.056	$[-0.167, 0.056]$	3/1
C3–C2	-0.028	$[-0.083, 0.000]$	1/0
C3–C0	-0.750	$[-0.944, -0.528]$	27/0

Table 4. FAR/clean acceptance by frozen reviewer configuration. Each cell has 18 faulty and 18 clean cases.

Reviewer	C0	C1	C2	C3
5.6-sol/max	.778/1.00	.000/1.00	.000/1.00	.000/1.00
5.4-mini/low	.778/1.00	.222/1.00	.111/.944	.056/1.00

$[-94.4, -52.8]$; 27 improving and zero worsening faulty decisions), while clean acceptance was unchanged. These large frozen-benchmark differences should not be read as population effects because the cases are generated and calls share context.

The decomposition is more diagnostic than the headline. C2 versus C1 changed FAR by -5.6 points (interval $[-16.7, 5.6]$; three improvements, one regression) and clean acceptance by -2.8 points. C3 versus C2 changed FAR by only -2.8 points (interval $[-8.3, 0]$; one improvement, zero regressions) and clean acceptance by $+2.8$ points. The sole faulty C2-to-C3 correction was a stale-artifact decision; an execution-hash-mismatch decision remained accepted under C3. All eight reviewer–case rows in the four semantic/out-of-scope lineages were already rejected in both C2 and C3, so this fixture supplies no evidence for broad semantic coverage or a stable addressability interaction.

Reviewer heterogeneity. Table 4 shows that the higher-compute reviewer rejected every faulty package once any artifact representation was visible, whereas the low-effort configuration improved stepwise and supplied every nonzero C1–C3 FAR. The single C2 clean rejection and its reversal at C3 also occurred in the low-effort configuration. Thus the aggregate C3–C2 change is not an independently replicated effect across reviewers. The exact McNemar values for C1–C0, C2–C1, C3–C2, and C3–C0 are 1.19×10^{-7}, .625, 1.0, and 1.49×10^{-8}, respectively; only the case-clustered intervals reflect within-case reviewer dependence.

Fault strata. Table 5 localizes the aggregate result. C0 accepted every mechanical stratum except claim-ceiling faults, but it already rejected all unsupported-mechanism rows and half the citation rows. Raw artifacts eliminated most remaining errors; negative-result omission was the largest C1 residual. Structured provenance corrected those two omissions but did not change stale-artifact or execution-hash decisions. C3 corrected one stale-artifact row and left one execution-hash row accepted. Because the semantic strata had zero acceptance in both C2 and C3, the planned addressable-versus-semantic incremental contrast is sparse and uninformative rather than evidence that the gate covers semantics.

Decision-level audit. Three low-effort-reviewer discordances illustrate what the aggregate cannot. For faulty case 004, C2 accepted a stale result because the claim matched its positive estimate; C3 rejected it after the report marked overall integrity as failed. For faulty case 018, C3 still accepted a result whose code hash disagreed with the execution hash, explicitly judging the integrity failure "unrelated" to the bounded claim. The report therefore informed but did not control the endpoint. In the opposite direction, clean case 031 was rejected under C2 because the reviewer asserted that train/evaluation disjointness was false even though the frozen execution recorded it as true. C3 exposed a PASS report and the same reviewer accepted the case. These examples show both possible diagnostic value and presentation sensitivity: a report may focus attention on a real stale edge, be discounted as immaterial, or counter a reviewer-invented inconsistency. They are post-hoc illustrations, not independent mechanism tests.

Costs do not show a simple monotone trade-off. C1 was 0.519 seconds per item faster than C0 despite more input tokens, whereas C2 added 0.254 seconds over C1. C3 used the most input tokens (1,266 per item) but was 0.123 seconds faster than C2. These are realized provider latencies, not controlled hardware measurements.

4. Discussion

The strongest result is not that a deterministic gate "solves" research review. It is that opening raw artifacts changed decisions far more than adding successively structured evidence. This supports evaluating provenance systems with component ablations: comparing only C0 and C3 would attribute a 75-point difference to the full interface even though 66.7 points appeared at C1. The small C3–C2 estimate is compatible with no incremental effect on this census.

TRACE-GATE nevertheless offers two practical design principles. First, a research agent should emit explicit claim–result–execution links and deterministic identity checks so reviewers can locate contradictions. Second, the checker output should remain non-directive until enforcement is separately evaluated; a forced veto confounds diagnostic evidence with policy. Semantic validity, causal interpretation, and citation entailment remain reviewer responsibilities.

Table 5. Accepted faulty decisions by stratum and condition (accepted/available). Each two-partner stratum has four reviewer decisions per condition; one-partner strata have two.

Fault	Scope	n	C0	C1	C2	C3	Fault	Scope	n	C0	C1	C2	C3
Result direction	M	4	4	0	0	0	Stale artifact	M	4	4	1	1	0
Metric mismatch	M	4	4	0	0	0	Execution hash	M	4	4	1	1	1
Contract drift	M	4	4	0	0	0	Negative omission	M	4	4	2	0	0
Code invariant	M	2	2	0	0	0	Claim ceiling	M	2	0	0	0	0
Mechanism claim	S	4	0	0	0	0	Citation entailment	S	4	2	0	0	0

Alternative explanations. The four conditions vary both information and presentation. C1 may help because contradictions become visible, while C2 may reduce navigation cost or merely make the package look more authoritative. C3 adds tokens, a salient PASS/FAIL vocabulary, and deterministic evidence simultaneously. The non-directive endpoint removes a mechanical veto but cannot remove automation bias. Fixed ordering within complementary shards improves pairing, yet condition-specific calls leave provider latency and cross-item context as possible explanations. The observed decomposition therefore identifies where decisions changed, not which cognitive mechanism caused the change.

Protocol deviation. The plan specified 16 calls without replacement. Four low-effort-reviewer attempts returned schema-invalid outputs without item IDs; they were excluded before inspecting verdicts, and successful planned batches were retained. Consequently, 20 provider attempts produced 16 analyzable batches. This protects the table from malformed rows but violates the frozen no-replacement criterion, so every condition effect here is exploratory rather than confirmatory.

5. Limitations

The benchmark is a deterministic generated fixture, not deployed peer review. It contains 18 lineages, two related model configurations, and one prompt/interface realization; natural faults may differ sharply. Items were separated from their paired counterparts but not from 17 other items in each call, so cross-item context and reviewer dependence remain. The gate taxonomy helped generate 14 addressable faults and therefore cannot test out-of-taxonomy generalization. Four semantic faults were easy for both structured conditions, preventing a meaningful boundary test. There were no human reviewers, no human-time outcome, no independent call-level replication, and no causal, non-inferiority, or reviewer-population inference. Token allocation and latency division are bookkeeping approximations, and two cited 2026 systems are preprints. Finally, the retry deviation may select for schema-following outputs even though exclusion was outcome-blind.

6. Conclusion

On a frozen paired fixture, raw artifacts sharply reduced model-reviewer false acceptance while structured provenance and a non-directive gate report added small, uncertain increments. The result argues for auditable AI-scientist packages and, equally, for ablations that prevent an automatic checker from receiving credit for evidence access or a forced rejection policy. Larger natural-fault benchmarks and human review are required before treating provenance gates as reliability interventions.

Reproducibility Statement

The frozen analysis comprises 36 package instances, 288 typed decisions, post-review gold labels, condition-specific costs, exact discordant-pair counts, and a case-clustered bootstrap with seed 1729 and 2,000 draws. No accelerator, human adjudication, or automatic C3 veto was used.

Impact Statement

Trace checks may improve research auditing, but false confidence in automated PASS labels could suppress necessary scientific review. We therefore expose the gate as non-directive evidence, report clean acceptance, and reserve semantic judgments for future human-centered evaluation.

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